Catering Staff Management System

Project Report — Production Build

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Prepared for: Badusha and Murshid's Event Management Company
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Version: 4.0 — Draft (1 open item in Section 14)

> **Core Properties — Stability · Performance · Security**
> Everything below is designed around these three first. Anything that doesn't serve them is deferred, not built.

Build approach: ship a simple, working v1 now. Add complexity — payments, radius search — only after the system has run for months and the team has adapted, and only in ways that never break or lose existing data. High-load days (Section 12) are handled by temporarily scaling infrastructure, not by permanently over-building the system.

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Appendix — Flow at a Glance

1. Roles & Permissions

Admin side

Role	Responsibility
Main Admin	Full control. Approves Sub Admin & Office Staff device access (Section 10). Future — once a payment gateway ships: sees financial/profit tracking.
Sub Admin	Oversees Office Staff activity. Monitoring/reporting access; day-to-day execution stays with Office Staff.
Office Staff	Verifies registrations, creates and manages work listings, handles daily operations.

Field side (low → high)

D Boy → C Boy → B Boy → A Boy → Supervisor → Captain → Site Manager

- A Boy and below can apply for work themselves.
- Supervisor and above never self-apply — they see listed work, and Office Staff contacts or assigns them directly.
- Treat this list as configurable, not hardcoded — positions are expected to change over time.

2. Registration

Personal information

- Full Name * (as on Aadhaar)
- Mobile Number *
- District * (dropdown)
- PIN Code *
- City *
- Home Address *

Location *(optional)*

- **Use Current Location** or **Select on Map**
- Shows detected address → user confirms **Yes** or taps **Change**
- Not required now, but capturing it sets up the future radius-search feature (Section 13) without needing to re-contact everyone later.

Profile photo

- Recommended at registration, can be completed later from Profile.

Identity

- Aadhaar Photo — Front and Back *
- Emergency Contact Number + Relation (Father / Mother / Brother) *
- Retention policy shown right here at upload, plainly — never hidden. Exact policy is still pending (Section 14); whatever it ends up being, the user sees it at this step.

Consent *(all required)*

- ☑ Uniform Instructions
- ☑ Company Rules
- ☑ Safety & Guidance
- ☑ Company policies, general

(Office Staff/Admin to supply the actual policy text these link to.)

Account creation
Send OTP → Enter OTP → Verify → Account Created

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3. Verification

Office Staff reviews new registrations.

- **Approved:** account activates; user is notified in-app + SMS; can now see and apply for work.
- **Rejected:** Office Staff attaches a reason (e.g. "Aadhaar photo is blurry — reupload"); user edits and resubmits; returns to the queue.

An account isn't usable for work until it's verified.

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4. Role Promotions

One-time migration (rollout only): every existing worker is registered as D Boy, then Office Staff manually sets each person's real current position. This happens once, while bringing the existing workforce onto the system.

"Completed work" = attended on-site, and the work has since been marked Finished by Office Staff or Site Manager (Section 5). No payment step exists yet, so attendance + Finished status is the whole test.

Steady state, from then on:

Step	Trigger	Type
Register → D Boy	Immediate	Automatic
D Boy → C Boy	3rd completed work	Automatic
C Boy → B Boy	8th completed work	Automatic
B Boy → A Boy	Performance nomination — see below	Manual
A Boy → Supervisor / Captain / Site Manager	Main Admin, Sub Admin, or Office Staff	Manual

A Boy is the equivalent of today's "Main Boy" role.

A Boy promotion — performance nomination, not a count:

- From a work's on-site staff list, a Site Manager or Captain can send a promotion request for any D/C/B Boy who stood out. Nothing to do with work count or tenure — a pure performance call.
- Requests land in Office Staff's **Promotion Requests** queue as a name card, showing who requested it.
- The same Boy can be requested again later — same or different work, same or different requester — the card just adds a counter (2×, 3×…). It's a signal, not a trigger.
- Office Staff opens the card, checks the Boy's profile, and manually promotes — or doesn't. No deadline, no cap on how many A Boys can exist, no waiting list.

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5. Work: Creation, Listing, Applying & Cancellation

Creating & listing work

- Office Staff lists a work. Default: open to all eligible users.
- Filters: District, PIN Code dropdowns (Radius — future, Section 13).
- Number needed per tier (D/C/B/A Boy, and separately Supervisor+) is optional — leave blank for open/all, or set a target.
- If a tier's number is set, applicants beyond it land on a substitute list, first-come-first-served.
- Supervisor and above never self-apply — Office Staff assigns or contacts them.

Applying (D/C/B/A Boy)

- First come, first served: Confirmed → Substitute → Full.
- Can't self-apply to a second listing on the same date — regardless of meal type.

Office override: same-day double-booking

- Office Staff can still assign someone to a second same-day work. Example: a breakfast/lunch work is running at a site, and a same-day dinner work nearby is short-staffed. Office Staff calls the Site Manager on the running site, who picks the people needed and sends back their names + numbers. Office Staff assigns them directly to the dinner work.
- This bypasses the same-day rule above — it's an assignment, not a self-apply.
- Tracked as its own assignment reason, separate from filling an unanswered substitute slot — it's the only case exempt from Section 7's late tracking, since arriving late from a same-day double-booking isn't on the worker.

Cancellation

- No self-cancel in-app. Staff calls the office; Office Staff removes them from the confirmed list.
- All substitutes for that listing are notified instantly (push + SMS); first to accept gets the slot.
- Office Staff can manually assign a substitute if no one responds in time.
- If a substitute has since confirmed elsewhere the same date, they're auto-dropped from this substitute list.

Status

`Created → Coming → Ongoing → Finished` — Coming/Ongoing change automatically by date/time; Office Staff or Site Manager manually presses Finish. This also closes out attendance for that work, which feeds the promotion count in Section 4.

Note: this replaces the old percentage-based slot calculator (4% / 8% / 16% / 72%) with simpler optional manual numbers — confirmed.

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6. Notifications

Delivery: push is automatic (free, instant, to everyone relevant). SMS is manual — Office Staff triggers it, staged by PIN Code → District → neighbouring District, stopping once a listing is filled.

Notification	Visible to
Verification approved/rejected	The user only
New work posted	Eligible users (matching district/tier filters)
Work edited or cancelled	That work's registered users only
Reminder — 8–9pm the night before	That work's registered users only, sent automatically
Custom message from Office Staff	That work's registered users only, sent on demand
Position upgrade	The user only, for now — later, all users (toggle once the system feels stable)
Ban	**All users**, with reason, from day one

Promotions build positive momentum once visible; bans build awareness of what gets someone removed. Bans are public immediately; promotions go public later, once things have settled.

7. Attendance (On-Site)

- The Site Manager assigns who takes attendance for a work — can assign more than one person. Captains and Supervisors get priority, but other roles can be assigned too. Office Staff, Sub Admin, or Main Admin can also make this assignment.
- Whoever's assigned marks attendance from their phone via the listing's staff list.
- Tapping a name card shows address, phone, district/PIN, and whether they're also booked on another work that day.
- A name booked elsewhere that day shows with a red indicator — informational, not automatically a problem (see Section 5's office override).
- A Boy counter assignment on-site is a verbal, ground-level call — not tracked in the system.
- From this same staff list, Site Manager/Captain can send a promotion request for a standout D/C/B Boy (Section 4).

Late marking

- Each work has a site time (e.g. 9:00am). Arriving up to 45 minutes after that is on time — no prompt, nothing recorded.
- Marking attendance past that 45-minute window prompts the attendance-taker: **"[Name] is late?" — Yes / No.**
- **Yes** → that work entry gets a Late status. Minutes late (counted from the site time, not from the 45-minute cutoff) are added to the staff member's profile totals: *Total Late Work* (count) and *Total Late Minutes* (sum).
- **No** → the attendance time is still recorded, but no Late status and nothing added to the profile.
- Exception: staff added through Section 5's office override are exempt from this prompt entirely.

8. Dashboards by Role

Role	Sees
D/C/B/A Boy	Home (listed work), My Works, Notifications, Profile
Supervisor / Captain / Site Manager	Assigned work, staff list + attendance marking + promotion requests, Profile
Office Staff	Verification queue, work listings (create/edit/finish), staff directory, promotion requests
Sub Admin	Office Staff activity overview
Main Admin	Everything above, plus account & device management (Section 10)

9. Staff Directory & Filtering

Filter by District, PIN Code, or position. Each result is a card:

```
[photo]  Mohammed Irshad      ● A Boy
     9876543210          Malappuram
```

Tap a card for the full profile. Color-coding by position is a visual detail to finalize during design.

10. Account Security

Sub Admin and Office Staff accounts are device-locked:

- First login from a device requires Main Admin approval (sees device model, IP, timestamp).
- Login from a new/different device is blocked until approved.
- Main Admin can see every account's active device, terminate sessions, and view full login history.

Extended from the original spec: since Office Staff now hold the sensitive day-to-day access (verification, staff data), the same device-lock that was Sub-Admin-only now covers them too — confirmed.

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11. Technology & Hosting

Platform: Django, delivered as a PWA — installs to the home screen, no App Store involved. Full push notifications on Android and iOS 16.4+; older iPhones get SMS as backup.

Component	Tool	Cost
App + database server	DigitalOcean Droplet, 2GB RAM	₹1,000/mo
Aadhaar/photo storage	Cloudinary (free tier)	Free
Push notifications	Firebase Cloud Messaging	Free
SMS	MSG91	~₹0.20/SMS
Domain + SSL	Any registrar + Let's Encrypt	₹1,000/yr

Estimated yearly cost: ≈ ₹14,800–16,840, including 750–1,600 SMS/month.

Growth (5,000+ staff, heavier DB load, more storage) is handled by upgrading within the same DigitalOcean setup — no migration, no data loss.

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12. Scaling for High-Volume Registration Days

For rare mass-registration days (2,000–3,000 people, once or twice every 2 years) — scale up temporarily, then scale back down:

- **Server:** resize the droplet to more RAM/CPU a day or two ahead; downsize after.
- **OTP/SMS:** confirm MSG91's rate limit / burst capacity in advance for the expected volume — this is a more likely bottleneck than the server itself.
- **Network:** venue Wi-Fi/hotspot bandwidth is usually the real constraint for photo/Aadhaar uploads — compress images client-side, and confirm connectivity at the venue beforehand.
- **Duplicates:** enforce a unique mobile number per account, and flag duplicate Aadhaar uploads for manual review — important on a rushed high-volume day, since duplicate accounts would also skew automatic promotion counts.

This is a checklist for that specific day, not a change to the everyday architecture.

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13. Deferred to Future Phases

Not built now — only after the system has run stably for months and the team has adapted:

- **Payment gateway & financial/profit tracking** (Main Admin only). Build it as an additive module on its own tables when the time comes, so it never touches or risks the existing registration/work data.
- **Radius/circle search** — search by distance around a work location, using the location data optionally captured at registration (Section 2). District/PIN/City filtering covers this need for now.

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14. Open Items Needing Confirmation

Aadhaar retention isn't decided yet — either delete right after verification, or hold for a few months and auto-delete after that. That choice will be made later, closer to when it's actually needed, not now.

Already settled, regardless of which way that goes: the retention policy is shown to the user at the Aadhaar upload step (Section 2), never

hidden. Also still worth a quick legal/client check given India's DPDP Act, phased in through 2026–2027 (full enforcement by May 2027) — sensitive government ID data, collected at real scale. Not legal advice, just flagging it for a proper check.

Appendix — Flow at a Glance

```
REGISTER (D Boy, automatic)
   │
   ▼
OTP VERIFIED → "Pending Verification"
   │
   ├── Office Staff reviews
   │     ├── ✓ Approved → Active, can apply for work
   │     └── ✗ Rejected (reason given) → user fixes → resubmits
   │
   ▼
ACTIVE STAFF
   │
   ├── 3rd completed work  → auto C Boy
   ├── 8th completed work  → auto B Boy
   ├── Site Manager/Captain sends promotion request → Office Staff decides → A Boy
   └── Manual, above A Boy → Supervisor / Captain / Site Manager
   │
   ▼
Work listed (push to all, SMS waves if needed)
   → Applies → Confirmed / Substitute / Full
   → Reports on-site → attendance marked by whoever's assigned (Section 7)
   → Office Staff or Site Manager finishes the listing → attendance count updates → feeds promotion count
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Prepared by Irshad for Badusha and Murshid's Event Management Company — July 2026.
Sections 1–13 reflect agreed direction. Section 14's remaining item should be confirmed before this is treated as final.